

Vision-Based Assessment of Right-Hand Piano Posture Using Monocular RGB Video

Max Shi

Computer Science and Software Engineering
University of Canterbury
Christchurch, New Zealand
msh254@uclive.ac.nz

Richard Green

Computer Science and Software Engineering
University of Canterbury
Christchurch, New Zealand
richard.green@canterbury.ac.nz

Abstract—This paper proposes a method for assessing right-hand piano posture from a single monocular RGB camera positioned in a diagonal front-view above the keyboard, requiring no depth sensor or specialised hardware. A 21-landmark hand estimator is used to extract per-finger proximal and distal joint angles, wrist pitch, mean MCP arch angle, and a normalised openness metric from each video frame. These features are mapped into discrete posture categories, curved, flat, collapsed, and neutral for each finger; low, neutral, and high for the wrist; and good, collapsed, or excessive for the hand arch. These results are then aggregated into a scalar score between 0 and 100 using a weighted rule-based classifier. A closed-hand override based on the openness metric is introduced to reduce false high scores caused by fingertip occlusion. Experiments across six controlled posture conditions show that the system distinguishes good posture (mean score 80.2) from low, flat, and high wrist conditions (mean scores 28.4–52.0). However, heavily occluded postures, such as *on nails* and *on knuckles*, produce high score variability, revealing a key limitation of RGB-only distal joint estimation. These results indicate that a low-cost webcam-based posture module is a feasible building block for an interactive piano tutoring system, though further work on multi-subject validation and occlusion handling is needed.

I. INTRODUCTION

Playing the piano safely and efficiently requires not only the ability to play the correct note at the correct time, but also maintaining healthy hand posture over extended periods of practice. Traditional piano pedagogy holds that a safe hand posture is characterised by appropriately curved fingers, a stable wrist, and an arched hand shape [5]. Modern teaching practices rely on the teacher to assist students in person in achieving the correct hand posture, something students without in-person access may struggle with. Many students practice alone most of the time and may receive only intermittent feedback on their posture, increasing the risk of musculoskeletal disorders related to playing [11].

Recently, it has been seen that consumer cameras, laptops, and mobile devices are prevalent in practice environments. This raises the question of whether a standard webcam placed near the piano could provide visual feedback on students' posture. Existing computer-vision systems for piano learning focus on timing, key detection, or fingering, and depth-based

posture tools require specialised hardware and only provide coarse, whole-hand labels. None of these approaches provides a low-cost, posture-focused tutor.

Thus, this paper proposes a method to investigate whether a system can be developed that uses only a monocular RGB camera in a home environment to provide detailed, meaningful posture feedback. Specifically, the question is whether a real-time hand landmark model can be used to estimate finger curvature, wrist height, and hand arch in a manner sufficiently useful to distinguish between good and poor piano posture.

Hence, this paper proposes a method for assessing right-hand piano posture from a monocular RGB video camera captured at a 'diagonal front view' setting, above the keyboard. Using this fixed setting and focusing on just one hand (the right) will make it easier to focus purely on posture, without the problem of mapping fingers to keys. The proposed method aims to provide real-time estimates of posture metrics with visual feedback, suitable for an interactive tutoring system.

II. BACKGROUND

A. Existing Work in Vision-Based Analysis

Lee *et al.* introduced a two-stream convolutional neural network approach that combined audio and video information to detect which keys were pressed and which fingering was used [1]. This approach achieved over 85% accuracy in pressed-key detection during Hanon exercises, with a mean average precision of 0.9 for finger identification, assuming the key locations were known. Although this may seem good, this approach requires a top-down approach and MIDI information.

However, this approach will limit the system's representation to bounding boxes for the fingers and keys; it does not estimate joint angles or the wrist, making a posture assessment impossible. There is no way to obtain information about the fingering technique or whether it is ideal.

Johnson *et al.* present one of the few systems that specifically target piano hand posture directly, using a depth camera set up above the keyboard. Using this camera, they classify each hand via the depth map as correct posture, low wrists, or flat hands [10], as seen in figure 1. Their pipeline segments

the left and right hands using per-pixel classification on depth images, then applies support vector machines over histogram-of-oriented-gradient and surface-normal descriptors to assign a posture label [10].

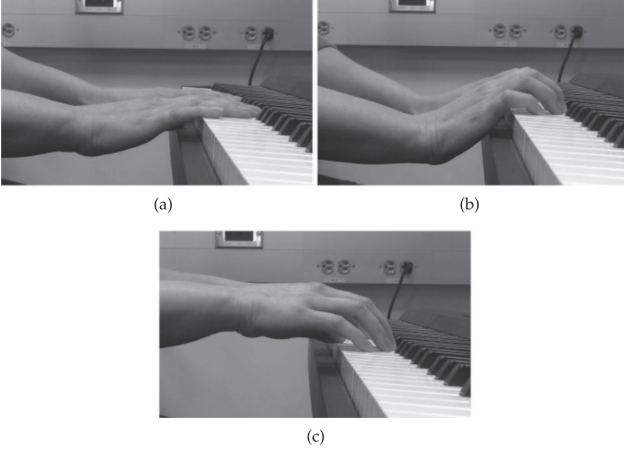


Fig. 1. Johnson *et al.*'s three posture classes. a) flat, b) low wrists, c) correct posture. Obtained from [10].

While this three-class classification is pedagogically useful and easy to visualise, it comes with several limitations. It treats the hand as a single unit and does not expose per-finger posture. As a result, the feedback may appear lacking in detail: the system can flag that a hand is “flat” or “low-wrist,” but it cannot provide detailed interpretations for each finger or track which fingers have the most inadequate posture [10].

Not only this, but the system's quality may suffer, as the evaluation is based solely on recordings of four beginner students in a controlled lab setting with the depth camera. In an environment with inadequate lighting or alternative camera placements, the system may produce inaccurate results. This may lead to repetitive practice of poor technique, potentially contributing to playing-related strain as seen in clinical and pedagogical studies [11].

Akbari and Cheng presented an image-processing pipeline based on edge detection and Hough transforms in order to locate piano keys and detect key motion in real time [2]. This method relied on a camera mounted above the keyboard with relatively stable lighting. The problem with this approach, however, is the requirement of stable lighting, which many casual home setups may not be able to meet.

B. Hand Posture

There is also research on hand posture; however, Gorodnichy investigated pianists' recognition of finger and hand configurations, demonstrating that fingertip position can be used to track hand configuration [3]. Additionally, Zeng and Sahagun proposed a method for analysing finger posture by extracting finger metrics from images and comparing them against predefined thresholds [4]. This system used geometric descriptors, such as angles at finger joints, to classify the

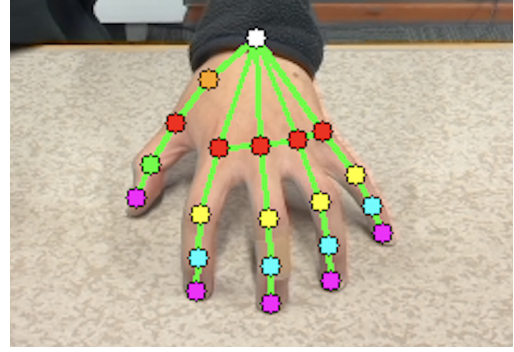


Fig. 2. The 21 MediaPipe Hand landmarks overlaid on the detected hand.

finger as curved, flat, or collapsed, which contributed to the posture score. Importantly, this work confirmed the ability to use geometric features and descriptors in order to capture posture.

From a broader hand-modelling perspective, Lee and Kunii emphasise that realistic hand motion cannot be obtained from unconstrained skeletal models; the human hand has around 27 degrees of freedom, and models that omit constraints may produce visually ‘valid’ but physically impossible results. The key takeaway here is that a posture analyser should not treat fingers merely as hinges; there should be constraints on how finger joint angles vary across the hand [12].

III. PROPOSED METHOD

A. Camera Configuration And Data Acquisition

The proposed method will use a single RGB camera positioned in front of the pianist, above hand height, so that the right hand is near the centre of the image, with the keyboard still visible. The camera will be tilted downwards towards the keyboard, providing an angled view of the back of the hand and wrist. For this method, a laptop webcam or phone camera of at least 30 fps with a resolution of at least 640×480 should be sufficient for real-time processing.

Each frame is processed by a hand tracker built on MediaPipe Hands [6], configured to detect up to two hands, but only the right hand. MediaPipe implements a two-stage pipeline: a palm detector that localises the hand region and a landmark regression network that predicts the 21-joint skeletal configuration for the given hand [6], [7].

Before defining the geometric features, it is useful to briefly introduce the anatomical terminology for the hand joints. Each non-thumb finger consists of a metacarpophalangeal (MCP) joint at the base, a proximal interphalangeal (PIP) joint in the middle, and a distal interphalangeal (DIP) joint near the fingertip, while the thumb has a carpometacarpal (CMC), MCP, and interphalangeal (IP) joint [9], [14]. Fig. 3 illustrates these joints and the corresponding bone segments.

B. Per-Finger Geometric Features

From the 21 landmarks, the system will compute joint angles for each non-thumb finger $f \in \{I, M, R, P\}$ (index, middle, ring, pinky) using a single primitive

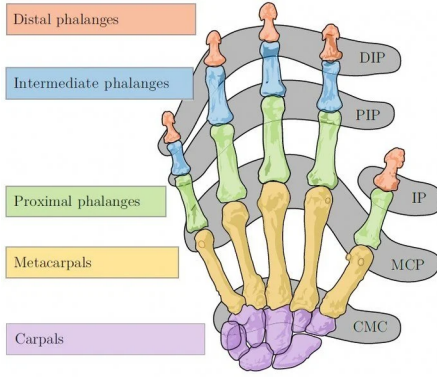


Fig. 3. Bones and joints of the human hand, showing the MCP (metacarpophalangeal), PIP (proximal interphalangeal), DIP (distal interphalangeal), IP (interphalangeal), and CMC (carpometacarpal) joints. Adapted from [18].

$$\theta(a, b, c) = \arccos \left(\frac{(a - b) \cdot (c - b)}{\|a - b\| \|c - b\|} \right), \quad (1)$$

with $\theta = 180^\circ$ corresponding to a fully straight joint. This is a standard joint-angle calculation which is used in many hand-pose and gesture-recognition systems which operate using 3D landmarks [8], [9].

Then, we can let $(P_{MCP}^f, P_{PIP}^f, P_{DIP}^f, P_{TIP}^f)$ denote the *world-space* positions of the four joints for finger f . Thus, the per-finger angles then become

$$\theta_{PIP}^f = \theta(P_{MCP}^f, P_{PIP}^f, P_{DIP}^f), \quad (2)$$

$$\theta_{DIP}^f = \theta(P_{PIP}^f, P_{DIP}^f, P_{TIP}^f), \quad (3)$$

$$\theta_{MCP}^f = \theta(P_{wrist}, P_{MCP}^f, P_{PIP}^f). \quad (4)$$

A benefit of using world-space landmarks is that they are independent of camera zoom or small shifts, which are almost certain to occur in home environments. For the thumb, the angles will be computed at the IP and MCP joints, but the thumb MCP is not included in the global “arch” metric because its flexion axis is not aligned with the non-thumb metacarpals.

Through the landmarks, two hand features are derived: the wrist pitch and overall hand arch. Wrist pitch is defined in the y - z plane (sagittal plane for a pianist), as the angle between the wrist and the centroid of the four non-thumb MCPs

$$\bar{P}_{MCP} = \frac{1}{4} \sum_{f \neq T} P_{MCP}^f, \quad (5)$$

$$\phi_{wrist} = \text{atan2} \left(-(\bar{P}_{MCP} - P_{wrist})_y, |(\bar{P}_{MCP} - P_{wrist})_z| \right). \quad (6)$$

Through empirical observation, an angle of -12° is a healthy, slightly raised wrist, and -28° corresponds to a collapsed wrist. The formulation of these metrics is similar to that of existing work, such as Johnson [10].

The other hand metric obtained from landmark metrics is the overall hand arch of the pianist, calculated from the mean MCP angle

$$\bar{\theta}_{MCP} = \frac{1}{4} \sum_{f \neq T} \theta_{MCP}^f, \quad (7)$$

which captures the “dome” of the knuckles, a feature piano pedagogy emphasises [5], and is also conceptually similar to MCP-angle summaries, primarily used in constraint-based hand models [12].

Lastly, the “openness” metric is computed via the landmarks as

$$\bar{p}_{MCP} = \frac{1}{4} \sum_{f \neq T} p_{MCP}^f, \quad s_{palm} = \|p_{mid_mcp} - p_{wrist}\|, \quad (8)$$

$$\Omega = \frac{1}{4 s_{palm}} \sum_{f \in \{T, M, R, P\}} \|p_{TIP}^f - \bar{p}_{MCP}\|, \quad (9)$$

where $\bar{p}_{MCP}$ is the 2D centroid of the four fingers (thumb not included) MCPs and p_{mid_mcp} is the middle finger MCP. Open hands would yield a value of $\Omega \approx 1.7$ – 2.3 in this setup, whereas a clenched fist would give a value of $\Omega \approx 0.6$ – 0.9 . A threshold is then chosen between these two values in order to detect when the hand is closed. By design, Ω ignores depth and remains robust to world-depth drift under heavy occlusion, a known issue in RGB-based 3D hand estimation [7], [9].

C. Classification

Posture calculation occurs in three stages: the per-finger posture calculation, the whole-hand wrist and arch calculations, and the closed-hand override, calculated from openness.

Per-finger posture is calculated via the PIP and DIP angles. Each non-thumb finger is assigned a discrete PIP posture label,

$$\pi^f \in \{\text{CURVED}, \text{NEUTRAL}, \text{COLLAPSED}, \text{FLAT}\}$$

using fixed-angle bands, which were initialised from piano pedagogy texts, and refined against labelled frames. Concretely, for each non-thumb finger f :

$$\pi^f = \begin{cases} \text{FLAT}, & \theta_{PIP}^f \geq \tau_{\text{flat}}, \\ \text{COLLAPSED}, & \tau_{\text{collapsed}} \leq \theta_{PIP}^f < \tau_{\text{flat}}, \\ \text{NEUTRAL}, & \tau_{\text{neutral}} \leq \theta_{PIP}^f < \tau_{\text{collapsed}}, \\ \text{CURVED}, & \theta_{PIP}^f < \tau_{\text{neutral}}, \end{cases} \quad (10)$$

This mirrors the angle-band strategy commonly used in piano-posture classification and in generic hand-pose recognisers based on geometric features [4], [8]. Additionally, a DIP override will force $\pi^f = \text{FLAT}$ when θ_{DIP}^f exceeds a per-finger threshold,

$$\text{if } \theta_{DIP}^f > \tau_{DIP}^f, \quad \pi^f \leftarrow \text{FLAT}, \quad (11)$$

such that flat fingertips are detected, especially when PIP angles are confused or ambiguous.

The hand, wrist, and arch categories are similarly mapped into “bands,” with wrist pitch ϕ_{wrist} mapped to

$$\omega \in \{\text{LOW}, \text{NEUTRAL}, \text{HIGH}\}$$

based on thresholds around an empirically chosen ideal angle, selected from previous pedagogical work [10]. Formally,

$$\omega = \begin{cases} \text{LOW}, & \phi_{\text{wrist}} < \phi^* - \delta_{\text{wrist}}, \\ \text{NEUTRAL}, & |\phi_{\text{wrist}} - \phi^*| \leq \delta_{\text{wrist}}, \\ \text{HIGH}, & \phi_{\text{wrist}} > \phi^* + \delta_{\text{wrist}}, \end{cases} \quad (12)$$

where ϕ^* is the ideal wrist pitch and δ_{wrist} is the tolerance band.

The mean MCP angle $\bar{\theta}_{\text{MCP}}$ is simultaneously mapped into a hand-arch state

$$\mu \in \{\text{COLLAPSED}, \text{GOOD}, \text{EXCESSIVE}\},$$

with thresholds chosen such that “on knuckles” playing is classified into collapsed states, and over-clenched fists are classified as excessive:

$$\mu = \begin{cases} \text{COLLAPSED}, & \bar{\theta}_{\text{MCP}} \geq \tau_{\text{arch-collapsed}}, \\ \text{GOOD}, & \tau_{\text{arch-excessive}} < \bar{\theta}_{\text{MCP}} < \tau_{\text{arch-collapsed}}, \\ \text{EXCESSIVE}, & \bar{\theta}_{\text{MCP}} \leq \tau_{\text{arch-excessive}}. \end{cases} \quad (13)$$

An additional closed-hand override was also implemented, determined by the “openness” of the hand. Due to the camera angle, when the DIP was occluded by the hand — a common scenario when playing very curved or on the knuckles — the classifier could not determine the angles correctly and hence assigned a good score to incorrect posture. The closed-hand override solves this problem by computing an openness metric that lowers the score when the hand is deemed closed. Concretely,

$$\text{if } \Omega < \tau_{\text{open}}, \quad \pi^f \leftarrow \text{COLLAPSED } \forall f, \quad \mu \leftarrow \text{COLLAPSED}, \quad (14)$$

where τ_{open} is the openness cutoff.

All of these thresholds are stored as configuration parameters, calibrated against a small dataset of frames captured with deliberate posture, following the process recommended by Johnson et al. for depth-based piano posture [10] and Zeng & Sahagun for mobile finger-posture analysis [4].

D. Scoring

The final scalar posture score S is a weighted sum of these three channels:

$$S = \underbrace{\sum_{f \in \mathcal{F}} w^{\text{PIP}}(\pi^f)}_{\text{Finger PIP}} + \underbrace{w^{\text{wrist}}(\omega)}_{\text{Wrist}} + \underbrace{\sum_{f \in \mathcal{F} \setminus \{\text{thumb}\}} \alpha_f w^{\text{MCP}}(\mu^f)}_{\text{MCP dome}}, \quad (15)$$

where $\mathcal{F} = \{\text{thumb}, \text{index}, \text{middle}, \text{ring}, \text{pinky}\}$, π^f and μ^f are the per-finger PIP and MCP posture labels, ω is the wrist category, and

$$\alpha = (1, 1, 1, 2)$$

for (index, middle, ring, pinky) emphasises the pedagogical importance of the pinky maintaining its arch. The lookup tables w^{PIP} , w^{wrist} , and w^{MCP} map posture categories to point values (e.g. CURVED $\rightarrow$ 10, NEUTRAL $\rightarrow$ 7, COLLAPSED $\rightarrow$ 3, FLAT $\rightarrow$ 0 for PIP), following the rule-based scoring style of previous posture work [4], [10]. Scores are normalised such that “ideal” posture yields approximately 100 points, and obviously faulty combinations yield much lower values.

Additionally, to reduce frame-to-frame jitter in scoring, the system maintains a buffer containing the last $N = 30$ scores and reports their mean as the “running” score, which is displayed to the user. This process smooths out the system’s noise while still allowing quick responses to changes in posture. Formally, if S_t is the raw score at frame t , the displayed score is

$$\bar{S}_t = \frac{1}{\min(t, N)} \sum_{k=\max(1, t-N+1)}^t S_k, \quad (16)$$

This type of window smoothing is very common in vision-based gesture systems [13].

IV. RESULTS

This section reports the results of the proposed posture analysis system on a small self-recorded test set. All recordings were captured using the same monocular RGB camera configuration described in the previous section, with the system operating in real time and producing per-frame posture metrics and a running score.

A. Implementation Details

The system was coded in Python, using the MediaPipe Tasks API for hand landmark detection, and OpenCV for video capture and visualisation. These experiments were run on a macOS laptop, with an Apple Silicon CPU and 16 GBs of RAM. Additionally, these experiments were created using Python 3.9, mediapipe 0.10.x, and opencv-python 4.13.x. The video capture was via the laptop webcam, configured at 30 fps and 1280×720 resolution. All recordings were used in the front-and-above camera position, with the camera facing the pianist, so that the full right hand and wrist are visible and remain in a consistent position.

B. Experimental Protocol

To evaluate the validity of the posture metrics and scoring system, an experiment was conducted across various hand configurations, collecting a small set of controlled recordings. For each posture condition, 5 10-second recordings were made, yielding approximately 300 posture-detection frames per recording. For each recording, the pianist played Exercise No. 1 from Hanon’s *The Virtuoso Pianist in 60 Exercises*, chosen because it spans a range of finger patterns and hand

TABLE I

SUMMARY OF POSTURE SCORES BY CONDITION, AVERAGED OVER FIVE 10 S RECORDINGS PER CONDITION. MIN/MAX COLUMNS SHOW THE LOWEST AND HIGHEST RUNNING SCORES OBSERVED ACROSS ALL RECORDINGS, AND “CLOSED” IS THE PERCENTAGE OF FRAMES IN WHICH THE HAND WAS CLASSIFIED AS CLOSED.

Condition	Mean	Std. Dev.	Min / Max	Closed (%)
Good posture	80.2	9.2	54 / 100	0
Low wrist	51.7	9.4	26 / 90	0
High wrist	28.4	15.4	19 / 84	75
Flat hand	52.0	8.3	22 / 84	0
Very bent / on nails	58.7	30.3	19 / 100	61
On knuckles	59.2	29.1	19 / 100	55

positions while remaining simple enough not to confuse the algorithm [17].

The following posture conditions were tested:

- **Good posture:** textbook curved fingers, neutral wrist, and arched hand.
- **Low wrist:** visibly dropped wrist with otherwise similar finger curvature.
- **High wrist:** visibly raised wrist above the hand.
- **Flat hand:** flattened fingers and reduced curvature at the PIP joints.
- **Very bent / on nails:** exaggeratedly curved fingers playing near the fingertips.
- **On knuckles:** extremely bent fingers with the hand rotated onto the knuckles, a known difficult case for landmark-based posture analysis.

For each frame with successful posture detection, the system would log the raw posture score, per-finger PIP and DIP angles, hand arch measurements, openness, and the running score. For this simple experiment, the per-condition behaviour is summarised by aggregating across the five recordings per posture type, with the mean and standard deviation of the running score being reported.

C. Score Behaviour Across Posture Conditions

The results summarise the average posture score for each condition, obtained by averaging the per-frame running scores across all five recordings per condition. As expected, the *good posture* condition achieved the highest average score, while *low wrist*, *flat hand*, and especially *high wrist* produced substantially lower scores. The deliberately exaggerated *very bent / on nails* and *on knuckles* conditions produced intermediate mean scores, but with noticeably higher variability.

These results show that the proposed approach can broadly distinguish between healthy and visibly poor hand posture. The gap of scores between the *good posture* condition and the *low wrist*, *flat hand*, and *high wrist* conditions is large enough to be meaningful for feedback in tutoring environments.

D. Per-Condition Behaviour

The individual finger/thumb metrics indicated that the scoring system worked well. In the *good posture* recordings, the non-thumb fingers were classified as curved for nearly all

detected frames, and the hand arch was also classified as good for every frame. The openness value was also consistently high, around ($\Omega \approx 2.5$ – 2.7), with no frames being marked as ‘closed-hand’. This shows that the system can report stable results under the intended good posture.

In the *low-wrist* condition, the mean score dropped by about 30 points compared to the *good posture* condition, reflecting wrist-position penalties in the scoring function, as seen in equation 15. Additionally, the mean MCP angle increased to around 146° , and a handful of frames were classified as having a ‘collapsed arch’, which is due to the visual impression of a flatter hand shape due to the low wrist. Openness remained stable, and the hand was never classified as closed.

In the *flat-hand* condition, the system primarily responded through the per-finger posture labels. The middle, ring, and pinky fingers were classified as *flat* in the large majority of frames across recordings, and the mean scores (around 52) were clearly below those in the good posture condition. However, some fingers were still classified as curved, such as the thumb and index finger, an issue likely due to the angle at which the fingers were relative to the camera.

The *high wrist* condition produced the lowest mean score among the conditions and showed the greatest separation from the good posture case. However, this condition produced an interesting result: in the high-wrist recordings, the openness metric dropped to around 1.5, triggering the closed-hand override for roughly 75% of frames. As a result, many of the frames were labelled as *collapsed* for both the finger posture and hand arch, which lowered the score, even though the primary error should have been the wrist height.

In the *very bent / on nails* and *on knuckles* conditions, the scores showed the expected low scores but the highest variability, as indicated by the high standard deviation. In the recordings, the lowest score was 20, and the highest was 100. Looking more closely at these cases, this is a classic example of the aforementioned occlusion problem: even when the DIP and TIP landmarks are occluded and unreliable, the landmark estimator still returns plausible coordinates, even though the hand is in an invalid posture. In the *very bent* and *on knuckles* recordings, the openness metric often dropped enough to trigger the closed-hand override (around 55–61% of frames), which forced the fingers and arch to be collapsed, but otherwise, when the hand was relatively open, the posture scores rose unexpectedly high, even though the configuration was still poor.

V. CONCLUSION

This paper investigated whether a monocular RGB camera and a real-time hand-landmark model can provide meaningful feedback regarding piano posture. The initial results suggest that it is feasible across broad posture conditions but remains limited in certain scenarios, such as highly occluded postures.

Compared to prior research, the proposed system offers an interesting trade-off. Johnson *et al.* classified piano hand posture into three categories (correct posture, low wrists, flat hands) from a Kinect-style depth map, reporting that their best



Fig. 4. Example posture frames for three conditions: good posture (left), high wrist (centre), and on knuckles (right). Each frame shows the overlaid landmarks and running score.

combination of depth-context features and normal-vector histograms achieved high accuracy on four beginner students, but required per-student models and a depth sensor mounted above the keyboard [10]. In contrast, the proposed system operates with an ordinary off-the-shelf RGB webcam in an ‘above-and-in-front’ view and emphasises per-finger and arch metrics rather than a three-way label. Zeng and Sahagun reported that a mobile phone app using geometric thresholds over finger angles could reliably flag flat and collapsed fingers in static images, but noted that accuracy degraded when the phone was placed at extreme side angles, where fingers became occluded [4]. The behaviour observed here is consistent with this: angle-band classification over MediaPipe landmarks is effective for clearly visible fingers, but becomes unstable under strong flexion and occlusion.

Additionally, these results also align with MediaPipe Hands accuracy studies. Zhang *et al.* reported high precision for palm detection and accurate 21-keypoint localisation in typical (unobstructed) hand configurations [6]. Later work compared MediaPipe landmark estimates to a known motion capture standard, finding small but non-zero positional errors even in controlled conditions [15]. Taking this into the piano posture domain, this suggests that although landmark noise is small in many conditions, cumulative errors, especially at the DIP and TIP joints, can still lead to misclassifications when the hands are heavily flexed or occluded.



Fig. 5. Example frames illustrating the occlusion problem. Left: a normal curved-hand posture. Middle: an *on nails* posture where fingertips partly occluded. Right: an *on knuckles* posture with heavy fingertip occlusion, closed-hand override triggered.

A. Occlusion of Distal Joints

A key limitation in the system is the sensitivity to occlusion, especially at the distal joints (DIP). In the deliberately exaggerated *very bent* / *on nails* and *on knuckles* conditions, the DIP and TIP joints were often hidden by the proximal phalanges or by the hand itself. Despite this, the landmark model often returned plausible-looking coordinates, resulting in inconsistent geometric features and widely varying scores. To prevent this, the openness-based closed-hand override was implemented, which mitigated the problem in many frames by forcing the fingers and arch into a collapsed state when Ω dropped below a certain threshold, but did not fully prevent false high scores.

The impact of this occlusion can be seen in Fig. 5. In the ideal curved-hand posture (left), the fingertips and DIP joints are clearly visible, and the landmarks align perfectly with the true anatomy of the hand, supported by the perfect score and accuracy analysis. In the *on nails* posture (middle), the fingertips are slightly occluded, yet the openness metric still classifies the hand as open, and without the closed-hand override activated, the posture still receives a high score. The *on knuckles* posture (right) shows the closed-hand override working as intended: when there is excessive occlusion, the override is triggered, forcing the system to label the fingers and hand as collapsed.

These examples highlight an important challenge for RGB single-camera posture analysis. Without explicit occlusion handling or additional multi-view information, distal joint estimations can be overconfident in incorrect and problematic postures.

B. Future Research

The present work has several important limitations that also suggest clear directions for future research. First, the evaluation was conducted on a small, self-recorded dataset with a single pianist and a fixed camera configuration.

A natural next step is to collect a larger dataset with multiple players, diverse hand sizes, and varied keyboard heights and camera placements, to measure generalisation and tune thresholds more robustly. This would also make it possible to report quantitative metrics directly comparable to prior studies, such as per-class accuracy on the three posture labels used by Johnson *et al.* [10], or per-finger curvature detection rates similar to those reported by Zeng and Sahagun [4].

Second, the observed occlusion failures suggest that temporal and confidence-based reasoning should be incorporated into the posture classifier. Rather than treating each frame independently, future versions of the system could incorporate temporal smoothing of landmark positions, reject low-confidence landmarks, and explicitly detect and down-weight frames with severe self-occlusion. Integrating additional cues, such as multiple RGB views or a lightweight depth sensor above the keyboard, could also help disambiguate DIP/TIP positions in challenging postures, bringing some of the robustness benefits of depth-based systems [10] while preserving the convenience of a front-facing camera.

Finally, a more comprehensive validation against expert pedagogy and clinical measures of playing-related musculoskeletal risk would be valuable. Previous work has linked poor wrist and hand posture to increased strain and injury risk in pianists [11], [16]. Future studies could correlate the proposed posture score with expert human ratings, electromyography, or wearable-based load measurements, and longer-term intervention outcomes. This would help determine whether a simple, landmark-based scoring system can serve as a reliable proxy for the more complex concept of “healthy” piano technique in real teaching and practice environments.

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